

Design decisions put to the owner

Every decision here was a fork the evidence could not settle on its own: either it changed the architecture, or it traded cost against reliability. Each entry gives the question as it was asked, the options offered, what was chosen, and why the question existed at all.

Dates are 2026. The standing rule from 22 September — **safety and reliability before cost, size and convenience** — is what several of these answers turn on.

1. The harness connector

Asked: the specification (\$7, deviation 1) called for the OEM 94-way connector to be replaced by functionally-split sealed connectors plus an adapter harness. The physical part had since been identified from photographs of the machine. Keep the deviation, or put the OEM connector on the board?

Chosen: the OEM Bosch 94-way on the board, no adapter. Spec deviation 1 is recorded as reversed.

Consequence: the board-side land pattern is now load-bearing and has to come from a Bosch distributor. Neither part number returns a public catalogue hit.

2. Four part-selection findings, approved together

These four were presented as recommendations and approved with one "ok" on 22 September. Each one came out of choosing a part and discovering the drawn circuit could not accept it.

2a. Where the injector and EGR current comes from — POWER-PATH

The problem: every high-current load (injector hold ≈ 10 A, EGR up to 6.4 A stall, relay coils) draws through connector pin 21, a 20 A fuse, the EMI filter and a reverse-battery FET. The reverse-battery block sized that FET for 5 A, and the EMI filter block never stated a DC current at all. A 22 μ H inductor that does not saturate at 15–20 A is a large part. Meanwhile pins 04 and 06 are separate battery feeds, fused 10 A each in the loom, that this board uses only to measure voltage.

Options: move the injector and EGR supply onto pins 04/06 with their own protection, leaving pin 21 for the electronics; or keep everything on pin 21 and resize the fuse, filter and reverse FET for the full load,

with one contact carrying it.

Chosen: move the high-current loads to pins 04/06. Status: not yet implemented.

2b. Boost converter stability while cranking — BOOST-SLOPE

The problem: the injector boost controller (TPS40210-Q1) adds a fixed internal slope-compensation ramp. Its Equation 9 caps the current-sense resistor at $VDD \cdot L \cdot f_{SW} / (60 \cdot (V_{OUT} + V_D - V_{IN}))$, and 80% of that is the recommended value. The drawn 82 mΩ passes at 13.5 V (ceiling 99.7 mΩ) and **fails at the 6 V cranking corner** (ceiling 44.9 mΩ, so 1.83× over): the converter goes subharmonic exactly when the engine is being started. Worse, a 330 μH inductor that saturates above the 2.2 A current limit is not a part that exists in surface mount — the search found 330 μH parts rated 0.2–0.4 A — and the buildable direction, a smaller inductor, lowers the ceiling further (100 μH allows 30 mΩ).

Options: feed the controller's supply pin from the output side, which is TI's own remedy and steepens the ramp, and use 100 μH; or accept subharmonic operation while cranking.

Chosen: feed VDD from the clamped output side, 100 μH. Status: not yet implemented. The finding is pinned as a check.

2c. Sensor-supply protection — SENSOR-PTC

The problem: the requirement asked for a resettable fuse of "R25 = 3.0 Ω nominal, -30% spread". Real PTCs are not specified that way: Bourns MF-USMF020 gives a **range, 0.40–5.00 Ω**, and at its 0.40 Ω minimum it contributes 14% of the series resistance the fault-current ceiling assumed. Its hold current also halves from 0.20 A at 23 °C to 0.10 A at 85 °C, against a ≥200 mA requirement.

Chosen (first pass): a fixed series resistor sets the fault ceiling and the fuse only trips. This was later superseded — see decision 4.

2d. The 3.0 V and 3.3 V input clamps — LOWV-CLAMP

The problem: fifteen zeners clamp the analog, speed and discrete inputs. They were specified at 5 mA but sit at tens of microamps, where a sub-5 V zener's knee is soft. The datasheet allows 10 μA of leakage at only 1 V. On a discrete input, 10 μA through the ~28 kΩ source is 0.28 V against a logic-high margin of 0.36–0.41 V while cranking; on an analog input it is a signal-dependent error on the ADC. The simulation's idealised sharp-knee zener hid all of it.

Chosen: replace them with a low-leakage clamp to the 3.3 V rail (BAV199). Status: done for the analog and cam inputs; the discrete inputs needed a different answer — see decision 3.

3. The discrete switch inputs

Why it came back: BAV199 works where a clamp only conducts on a fault. The five switch inputs clamp **continuously**, because a 6–40 V input cannot be divided into logic range otherwise: reading a logic high at 6 V needs a divider ratio above 0.385, and staying under 3.3 V at 40 V needs one below 0.0825. Those do not overlap. Three candidate clamps were checked and all three fail:

Candidate	Why it fails
3.0 V zener (BZX84-C3V0-Q)	soft knee at μA ; the leakage that started this
Rail clamp (BAV199-Q to 3V3)	parks the pad at 3.9 V continuously — beyond the pin's rating, as an operating point rather than a fault
Shunt reference (LM4040-Q1 3.0 V)	draws $\sim 35\ \mu\text{A}$ of bias below breakdown (TI SNOS633N, Fig. 5-4); the cranking-dip input can supply only $20\ \mu\text{A}$, so the node sags to $\sim 2.6\ \text{V}$ with no guaranteed bound, and it fails outright on the pull-up channels

Options offered: an NPN buffer per input; a stronger drive plus the LM4040, with special cases for the two contact-to-ground channels; or a dedicated automotive switch-detection IC on SPI.

Chosen: the NPN buffer. $10\ \text{k}\Omega$ into the base, $4.7\ \text{k}\Omega$ and $4.7\ \mu\text{F}$ to ground, a BAV199-Q diode holding the base against reverse battery, and the transistor's collector as the MCU pin with a $10\ \text{k}\Omega$ pull-up to 3.3 V. The pad now only ever sees 0–3.3 V, and nothing on it has to conduct continuously.

What the safety rule changed inside this decision: the input resistor was first drawn at $100\ \text{k}\Omega$, which draws $0.11\ \text{mA}$ through a closed contact at 12 V. Non-gold switch and relay contacts at that current oxidise into an intermittent open — a field failure, not a bench one. $10\ \text{k}\Omega$ gives $1.1\ \text{mA}$. Everything is checked at the worst corner rather than typical: hFE 30 (datasheet minimum 60, halved for $-40\ ^\circ\text{C}$) still pulls the pad to 0.03 V from a 6 V input, and a typical part at $125\ ^\circ\text{C}$ stays off below a 1.3 V input.

Cost: the logic is inverted — pad low means input active — which is recorded in the pin map, and the failure modes (transistor open reads inactive, collector-emitter short reads active) are handed to the firmware plausibility spec, because one polarity cannot make every channel fail safe.

Status: done.

4. The 5 V sensor supply

Why it came back: the fix approved in 2c does not survive the real source impedance. The sensor rail block modelled the 5 V supply as ideal behind 0.1 Ω , but it comes from a 1 A buck. So:

- a shorted sensor harness hits the buck's current limit long before any fuse trips; 5V_MAIN collapses, 3V3_MCU follows it, and the supervisor shuts the engine down. One chafed wire stops the machine.
- a sensor supply wire shorted to **battery** back-feeds 24 V into 5V_MAIN through the fuse. No series resistor or PTC prevents that.

Options offered:

Option	What it buys	What it costs
Tracker per group + overvoltage cut-off	each group survives a short to ground <i>and</i> to battery; other groups and the MCU are unaffected	about 8 more parts
Series resistor + PTC (as approved in 2c)	cheapest	a harness short browns out the MCU; a battery short back-feeds the rail
Trackers fed directly, with a lower main TVS	fewer parts	rewrites the whole board's transient design, and needs the alternator question (U16) answered first

Chosen: a tracking regulator per sensor group (TPS7B4253-Q1, whose output survives -1 to $+45$ V and limits at 301–520 mA, tracking 5V_MAIN to ± 4 mV) behind a 100 V-class overvoltage cut-off.

Why the cut-off is needed: the tracker's input pin is rated 45 V and this board's battery rail reaches 73.3 V for 50 μ s on ISO 7637-2 pulse 2a. The cheap alternatives were worked through and discarded: a series resistor with a TVS must clamp below 45 V, so it would conduct through the entire 400 ms load dump (about 7 J), and any series resistance breaks the 0.52 V of dropout headroom at a 6 V cranking input; an emitter or source follower loses its own V_{be}/V_{gs} at that same corner. A cut-off instead disconnects for the 50 μ s pulse, while a 47 μ F capacitor holds the sensors up (0.16 V of droop), and stays connected through the 40 V dump.

Status: not yet implemented.

Where these came from

Nothing here was found by reading the schematic. Each one surfaced when a part number was chosen and its datasheet was checked against what the drawing assumed — which is the same shape as every earlier finding in this project: a document and an executable check disagreeing, with nothing comparing them until someone did.